Supplement: Carryover representation and inference for biomarker-treatment interaction tests in aggregated N-of-1 trials under covariance moderation

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Contents

This document is a companion to the manuscript "Carryover representation and inference for biomarker-treatment interaction tests in aggregated N-of-1 trials under covariance moderation: a factorial simulation comparison of nine analysis specifications" (pmsimstats team). The main manuscript was kept to a working length by moving six of its eleven Tier 2 sensitivity blocks here in full (S1, S4, S10, S11) or in part (S2, S8; the main text keeps each block's headline finding and a one-paragraph summary, with the full cell-by-cell detail below). A final section (S12) is exploratory material outside the main manuscript's stated scope: an Architecture A (mean moderation) counterpart to the main manuscript's Figures 1-3, included here for comparison only. Block, specification, and G-code notation match the main manuscript exactly; this supplement does not stand on its own and should be read alongside it.

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1 S1: within-factor autocorrelation

The main text (Section 3.6) summarizes this block's finding: the Exposure-weighted advantage is preserved at every autocorrelation level examined. Full detail follows.

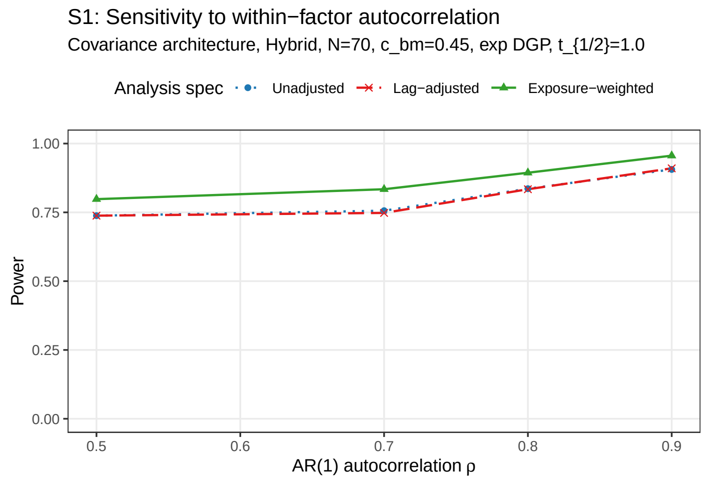

Figure 1: Power by analysis specification across AR(1) autocorrelation $\rho \in \{0.5, 0.7, 0.8, 0.9\}$.

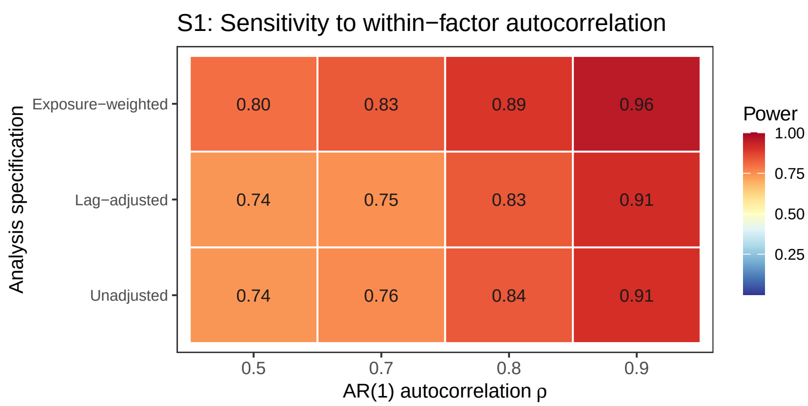

Figure 2: Block S1: power by AR(1) autocorrelation and analysis specification, same data as the line plot above.

All three specifications gain power monotonically with ρ, from 0.798 ± 0.018 (Exposure-weighted) and 0.738 ± 0.020 (the other two, identical to three decimals) at $\rho = 0.5$, to 0.956 ± 0.009 against 0.906 to 0.910 at $\rho = 0.9$. The Exposure-weighted advantage, roughly 0.05 to 0.08 in absolute power, is preserved at every autocorrelation level examined, suggesting autocorrelation strength and specification choice act additively rather than interacting. Lag-adjusted tracks Unadjusted throughout, never differing by more than 0.008.

2 S2: analyst-truth half-life mismatch, full detail

The main text (Section 3.6) reports this block's practical conclusion: an approximate half-life is adequate, with under-assuming safer than over-assuming. Full cell-by-cell detail follows.

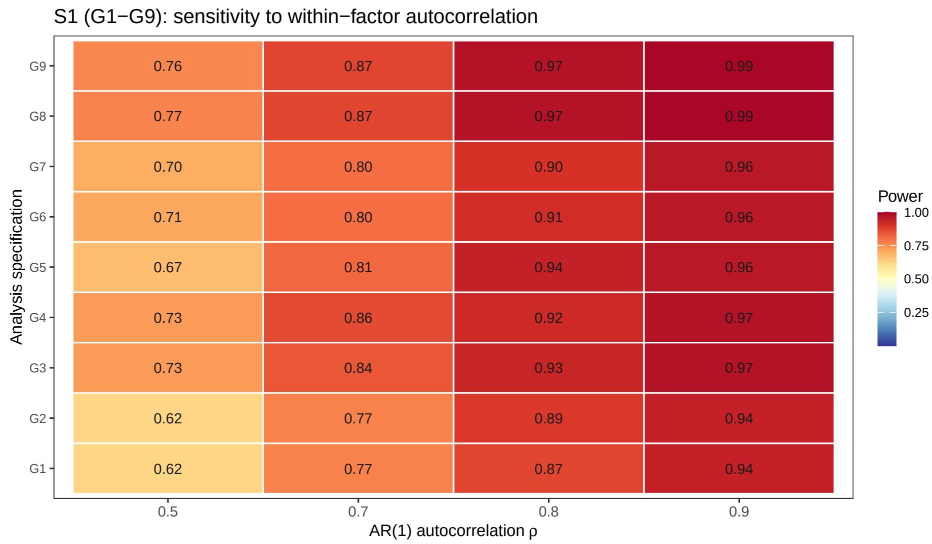

Figure 3: Block S1 extended to all nine G1-G9 specifications (preliminary $n_{sim} = 100$).

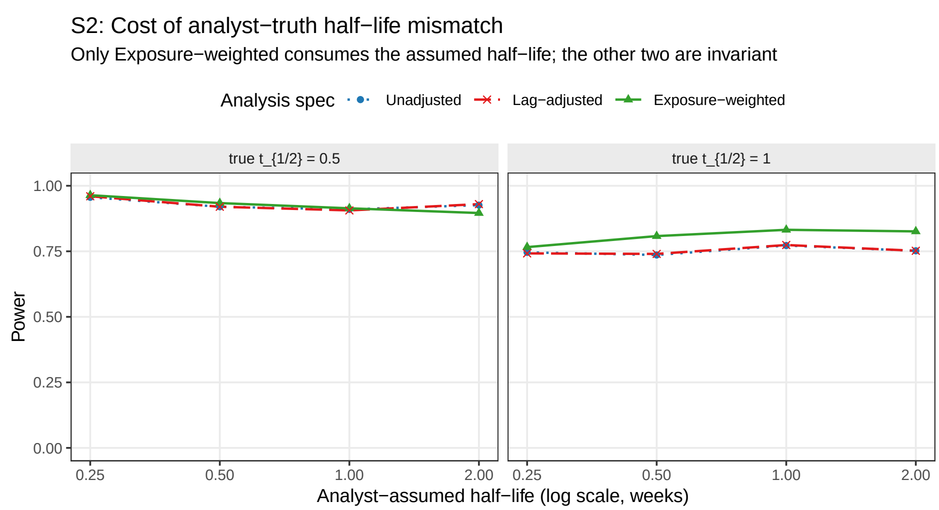

Figure 4: Power as the analyst's assumed half-life departs from the true DGP half-life. Only Exposure-weighted consumes the assumed half-life. Unadjusted and Lag-adjusted are invariant to it by construction and appear as flat reference lines.

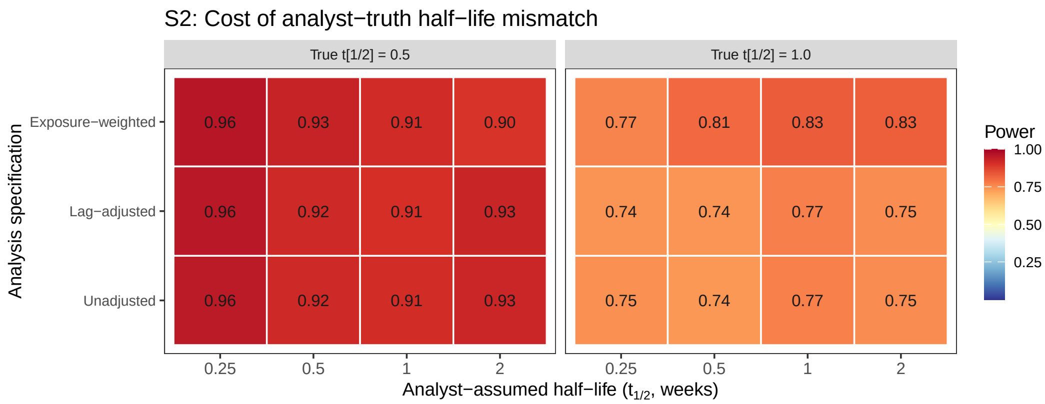

Figure 5: Block S2: power by analyst-assumed half-life and analysis specification, faceted by true half-life, same data as the line plot above.

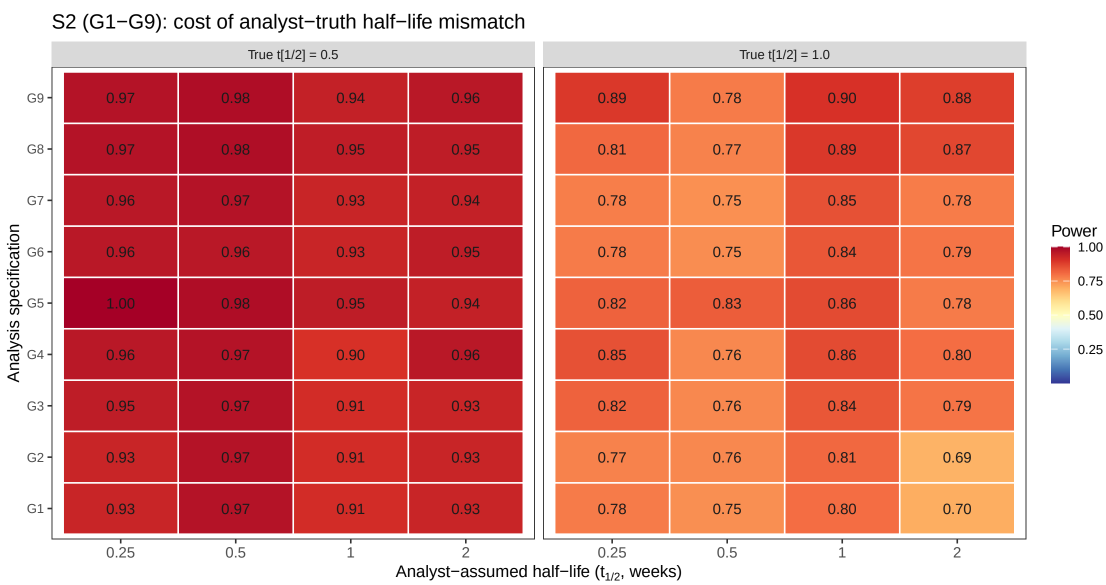

Figure 6: Block S2 extended to all nine G1-G9 specifications (preliminary $n_{sim} = 100$).

Block S2 crosses two true half-lives ($t_{1/2} \in \{0.5, 1.0\}$ weeks) with four analyst-assumed values ($\hat{t}_{1/2} \in \{0.25, 0.5, 1.0, 2.0\}$ weeks), giving eight cells. Only Exposure-weighted consumes $\hat{t}_{1/2}$, so the cost of a wrong assumption must be read within a true half-life rather than pooled across the block.

The penalty is modest. At a true half-life of 1.0 weeks, Exposure-weighted attains 0.832 correctly specified, versus 0.766, 0.808, and 0.826 at assumed values of 0.25, 0.5, and 2.0 weeks. The worst case, a fourfold under-assumption, costs 6.6 points and still leaves it above both alternatives (0.746, 0.742). At a true half-life of 0.5 weeks it attains 0.934 correctly specified, against 0.964, 0.914, and 0.896 at the same four assumed values. Because Unadjusted and Lag-adjusted cannot depend on $\hat{t}_{1/2}$, their variation along that axis is pure Monte Carlo error, ranging 0.910 to 0.960 at a true half-life of 0.5 weeks. This roughly four- to five-point noise floor explains the apparent anomaly that assuming 0.25 weeks (0.964) beats correct specification (0.934), a difference well inside it. It also qualifies the one case where the direction is worth noting. At a true half-life of 0.5 weeks with an assumed value of 2.0 weeks, Exposure-weighted falls to 0.896, below both alternatives (0.926, 0.930), so a fourfold over-assumption erodes its advantage to a tie.

The practical reading, given in the main text, follows from the limiting-case relationship of Section 2.4: shrinking $\hat{t}_{1/2}$ drives the predictor toward the binary indicator, which remains serviceable, whereas inflating it drives the predictor toward a constant, at which the on-off contrast and the estimand itself degenerate.

3 S4: biomarker-moderation effect-size curve

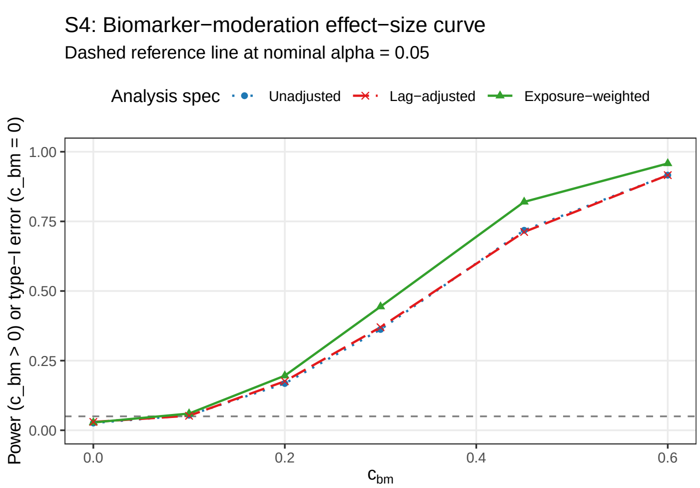

Figure 7: Power across the biomarker-moderation effect-size range $c_{bm} \in \{0, 0.10, 0.20, 0.30, 0.45, 0.60\}$. The $c_{bm} = 0$ point is the null for Type I verification.

The power curve is sigmoidal for all three specifications. At $c_{bm} = 0$, empirical Type I error sits below nominal for each (0.026 to 0.030). Power then rises through the mid-range and saturates once $c_{bm} \geq 0.45$, reaching 0.916 to 0.958 at $c_{bm} = 0.6$. The Exposure-weighted advantage over the unadjusted benchmark is largest at intermediate effect sizes, +0.082 absolute at $c_{bm} = 0.30$

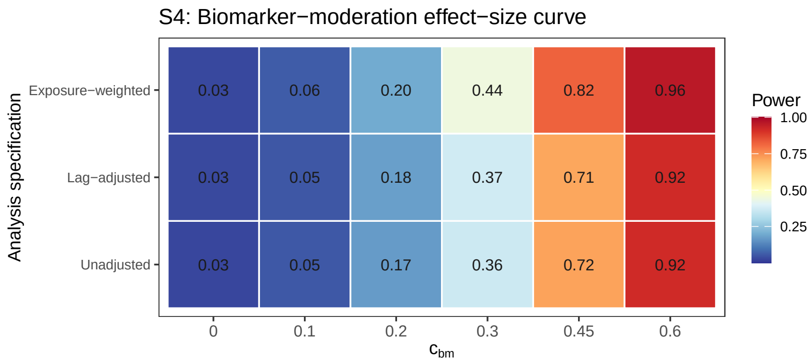

Figure 8: Block S4: power by biomarker-moderation effect size and analysis specification, same data as the line plot above.

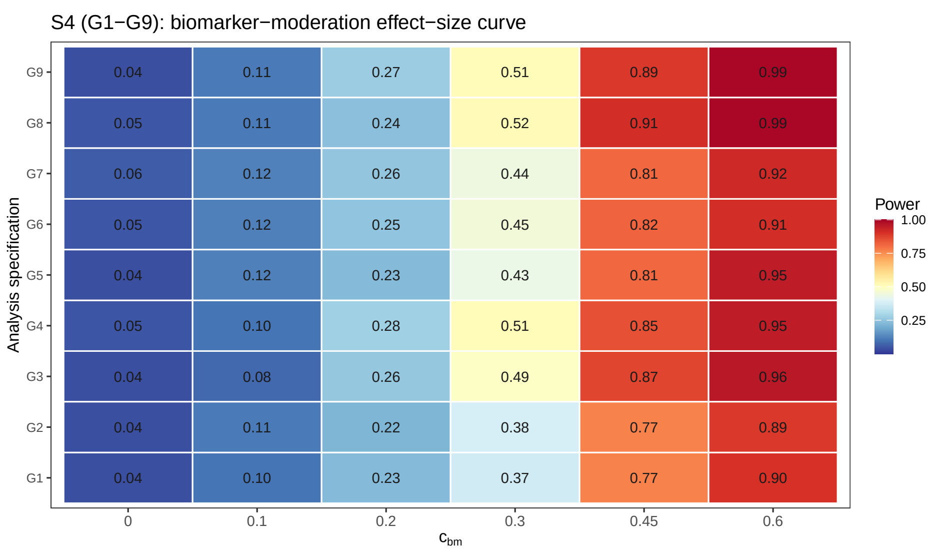

Figure 9: Block S4 extended to all nine G1-G9 specifications (preliminary $n_{sim} = 100$).

and $+0.102$ at $c_{bm} = 0.45$, shrinking to $+0.042$ at $c_{bm} = 0.60$ as all three approach the ceiling, so the advantage matters most for trials targeting a moderate biomarker-treatment interaction. Lag-adjusted and Unadjusted coincide across the entire range, differing by at most 0.008.

4 S8: architecture sensitivity, full detail

The main text (Section 3.6) reports this block's headline finding: `E7` and `E9` both beating the unadjusted benchmark at the Hybrid reference cell is architecture-specific, joining Exposure-weighted in a pattern that grows with carryover severity, and Lag-adjusted remains indistinguishable from Unadjusted under both architectures. Full cell-by-cell detail follows.

Every result reported in the main text, Tier 1 and blocks S1-S7 alike, is restricted to Architecture B (covariance moderation), the manuscript's stated scope (Section 2.3). Block S8 asks a narrower question than a full architecture comparison, which is the purpose of the companion manuscript on DGP architectures[1]: does the ranking among analysis specifications established under Architecture B hold under Architecture A (mean moderation) as well, or is the choice of carryover representation itself architecture-dependent? We refit the five specifications of block S7's table (`E1`, Lag-adjusted `E3`, Exposure-weighted `E2`, `E7`, `E9`) at the same reference cell under both architectures, at three half-lives ($t_{1/2} \in \{0, 0.5, 1.0\}$) rather than the single reference value, so the check covers whether the ranking is architecture-dependent uniformly or only once carryover is severe enough. Because c_{bm} is not calibrated to equal power across architectures, the two produce materially different absolute power at the same nominal value (Section 1;[1] shows the mean-moderation channel requires several times the nominal effect size for a comparable power loss under carryover); only the within-architecture ranking is the object of interest here.

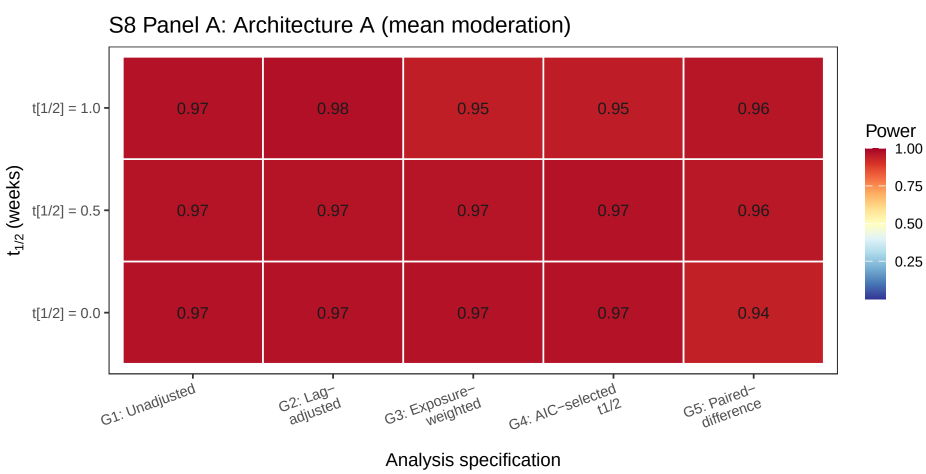

Figure 10: Block S8, Panel A (Architecture A, mean moderation): power by carryover half-life (rows) and analysis specification (columns). Same data as the Architecture A rows of the table above.

Table 1 and Figures 10/ 11 show the S7 result (`E7` and `E9` both beating `E1`) is architecture-specific, joining Exposure-weighted in a pattern that grows with carryover severity rather

Table 1: Block S8: analysis-specification ranking by DGP architecture and carryover half-life, at the reference cell (Hybrid, $N = 70$, $c_{bm} = 0.45$, exponential DGP, $n_{sim} = 500$). Absolute power is not comparable between architectures at the same nominal c_{bm}; only the within-architecture ranking is of interest.

Architecture	t1/2	Specification	Power (MCSE)	Non-conv
B (covariance)	0.0	Unadjusted	0.982 (0.006)	0.000
B (covariance)	0.0	Lag-adjusted	0.980 (0.006)	0.000
B (covariance)	0.0	Exposure-weighted	0.982 (0.006)	0.000
B (covariance)	0.0	AIC-selected t1/2	0.980 (0.006)	0.000
B (covariance)	0.0	Paired-difference	0.974 (0.007)	0.000
B (covariance)	0.5	Unadjusted	0.922 (0.012)	0.000
B (covariance)	0.5	Lag-adjusted	0.920 (0.012)	0.000
B (covariance)	0.5	Exposure-weighted	0.936 (0.011)	0.000
B (covariance)	0.5	AIC-selected t1/2	0.936 (0.011)	0.002
B (covariance)	0.5	Paired-difference	0.934 (0.011)	0.000
B (covariance)	1.0	Unadjusted	0.760 (0.019)	0.000
B (covariance)	1.0	Lag-adjusted	0.760 (0.019)	0.000
B (covariance)	1.0	Exposure-weighted	0.836 (0.017)	0.000
B (covariance)	1.0	AIC-selected t1/2	0.850 (0.016)	0.000
B (covariance)	1.0	Paired-difference	0.842 (0.016)	0.000
A (mean)	0.0	Unadjusted	0.972 (0.007)	0.000
A (mean)	0.0	Lag-adjusted	0.972 (0.007)	0.000
A (mean)	0.0	Exposure-weighted	0.972 (0.007)	0.000
A (mean)	0.0	AIC-selected t1/2	0.970 (0.008)	0.000
A (mean)	0.0	Paired-difference	0.942 (0.010)	0.000
A (mean)	0.5	Unadjusted	0.968 (0.008)	0.000
A (mean)	0.5	Lag-adjusted	0.968 (0.008)	0.000
A (mean)	0.5	Exposure-weighted	0.966 (0.008)	0.000
A (mean)	0.5	AIC-selected t1/2	0.970 (0.008)	0.002
A (mean)	0.5	Paired-difference	0.960 (0.009)	0.000
A (mean)	1.0	Unadjusted	0.972 (0.007)	0.000
A (mean)	1.0	Lag-adjusted	0.976 (0.007)	0.000
A (mean)	1.0	Exposure-weighted	0.946 (0.010)	0.000
A (mean)	1.0	AIC-selected t1/2	0.950 (0.010)	0.000
A (mean)	1.0	Paired-difference	0.964 (0.008)	0.000

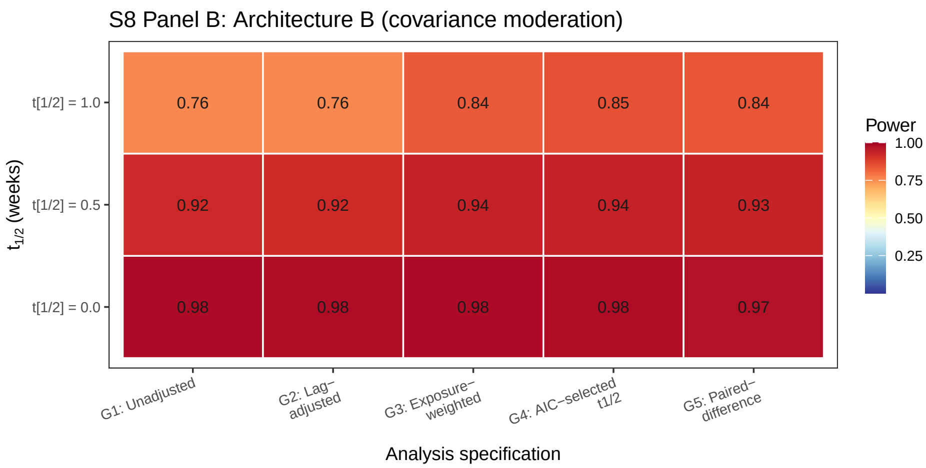

Figure 11: Block S8, Panel B (Architecture B, covariance moderation): power by carryover half-life (rows) and analysis specification (columns). Same data as the Architecture B rows of the table above. Not directly comparable to Panel A in absolute terms.

than switching on at some threshold. At $t_{1/2} = 0$ every specification collapses to nearly the same value in both architectures (no carryover to represent), so `E1`, `E2`, and `E7` agree there by construction (all 0.97-0.98 under B, 0.97 under A); `E9`'s slightly lower value even at $t_{1/2} = 0$ (0.974 under B, 0.942 under A) reflects sampling variance in its two-stage estimator rather than a carryover effect. Under Architecture B, the ranking at $t_{1/2} = 1.0$ is `E7` (0.850) $>$ `E9` (0.842) $>$ `E2` (0.836) $>$ `E1`/`E3` (0.760): both new specifications not only beat the unadjusted benchmark, as in block S7, but slightly exceed Exposure-weighted, the manuscript's own current recommendation. At $t_{1/2} = 0.5$ the same ordering holds but compressed (`E7` 0.936 essentially ties `E2` 0.936; `E9` 0.934 close behind; `E1`/`E3` 0.920-0.922 trail by about 1.5 points). Under Architecture A the pattern reverses for `E7` and `E2` alike. At $t_{1/2} = 1.0$, `E1`/`E3` lead (0.972/0.976), `E9` trails only slightly (0.964, a 0.8-point gap), while `E2` (0.946) and `E7` (0.950) trail more substantially (about 2.2 to 2.6 points, roughly 1.5 MCSE and so a real, if modest, reversal rather than pure noise). This matches the pattern already reported from the full-scope grid in the archived drafts, that Exposure-weighted is marginally dominated in every design examined under mean moderation (Section 4.6 of the main text); `E7` inherits that same architecture-sensitivity since it is built on Exposure-weighted's formula. `E9`, built on nothing but a raw on-drug/off-drug contrast, is comparatively more robust to architecture, small under B, smaller still under A, though never reversing to a large disadvantage either way. Lag-adjusted (`E3`) remains statistically indistinguishable from Unadjusted (`E1`) under both architectures at every half-life (largest gap 0.004, well under one MCSE), extending Section 3.3's clean null to Architecture A as well. The one specification recommendation the main manuscript makes, exposure weighting in the Hybrid and OL+BDC designs (Conclusions), is therefore itself conditional on Architecture B and grows more important as carryover worsens; the same qualification now extends to `E7`. `E9` is the one specification in this table whose advantage, where it exists, does not appear to depend on which architecture generated the data, though it was tested only at this one design and reference cell.

Table 2: Block S10: G1-G9 at the reference cell (Hybrid, $N = 70$, $c_{bm} = 0.45$, $t_{1/2} = 1.0$, exponential DGP, Architecture B, $n_{sim} = 500$). G1-G7 and G9 share common random numbers within this block; G8 is from the independent Block S6 run (different seed) and is not paired with the rest.

Code	Method	Power (MCSE)	Non-conv
G1	Unadjusted	0.748 (0.019)	0.000
G2	Lag-adjusted	0.748 (0.019)	0.000
G3	Exposure-weighted	0.826 (0.017)	0.000
G4	AIC-selected t1/2	0.832 (0.017)	0.000
G5	Paired-difference	0.804 (0.018)	0.000
G6	Unadjusted + CR2	0.776 (0.019)	0.010
G7	Lag-adjusted + CR2	0.764 (0.019)	0.016
G8	Exposure-weighted + CR2	0.863 (source: Block S6)	–
G9	AIC-selected t1/2 + CR2	0.870 (0.015)	0.000

5 S10: does a CR2 correction change the ranking among G1-G9?

The main text (Section 3.6) reports this block's headline finding: CR2 helps every specification it was applied to, and lifts `G4` enough that `G9` (`G4`+CR2) overtakes every other specification tested. Full detail follows.

Section 3.5 (Block S6) showed the model-based interaction test is mildly conservative and that a CR2 cluster-robust standard error recovers two to five points of power without disturbing the ranking among `E1`, `E2`, and `E3`. That correction had not been applied to `E7` or `E9`. This block asks two questions at once, adding a compact naming layer (`G1-G9`) for the resulting nine-way comparison: does CR2 help `E7` and `E1/E3` the way it already helps `E2` (Section 3.5), and does it change which specification leads at the Hybrid, Architecture B reference cell? `G1-G5` are the five specifications introduced so far (`E1`, `E3`, `E2`, `E7`, `E9`, respectively); `G6-G9` cross the CR2 correction with `G1`, `G2`, `G4`, and `G3`. `G8` (`E2`+CR2) already exists as Block S6's reference-cell result and is included here from that independent run (different seed, not sharing common random numbers with `G1-G7/G9`, so its comparison to the rest of this panel is not as tightly paired as the comparisons among `G1-G7/G9` are with each other). A tenth combination, `G5`+CR2, is not included: `E9`'s paired-difference regression collapses to one row per patient before fitting, so there is no repeated-measures cluster structure left for CR2 to correct.

Table 2 and Figure 12 answer both questions together. CR2 helps every specification it was applied to, consistent with Section 3.5: `G1/G2` (Unadjusted and Lag-adjusted, both 0.748 model-based) rise to `G6` 0.776 and `G7` 0.764, gains of 2.8 and 1.6 points respectively, in line with the two to five points Block S6 already documented for `E2`. The ranking among the five base specifications is unchanged by these particular gains, since `G1/G2` remain well below `G3-G5` regardless. What does change the ranking is `G9`. CR2 lifts `G4` (AIC-selected half-life, 0.832 model-based) to 0.870, a 3.8-point gain, larger than any other CR2 correction observed

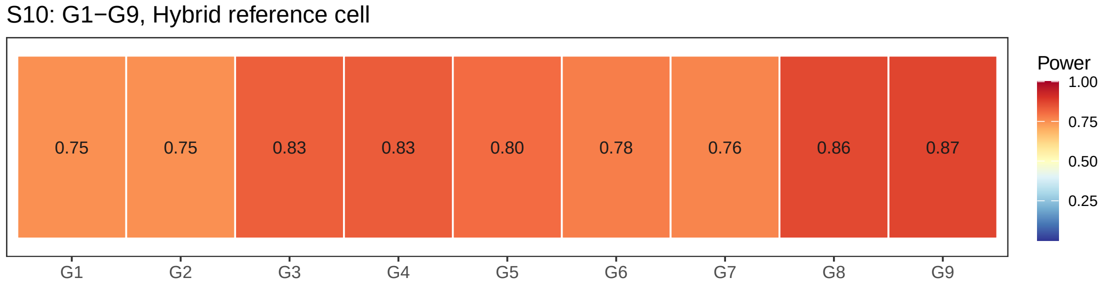

Figure 12: Block S10: power for G1 through G9 at the Hybrid, Architecture B reference cell, same figures as the table above.

here or in Block S6, and enough for `G9` to overtake every other specification tested, including `G8` (0.863, the Block S6 reference-cell result for Exposure-weighted+CR2). `G9`'s gain is also notable for what did not happen: `E7`'s AIC-selection step raised a Type I error concern earlier in this manuscript's development (post-selection inference, since the same data are used to choose the half-life and then test the resulting coefficient), and one might have expected a CR2 correction, which widens standard errors, to partly offset an anti-conservative test by coincidence. Instead every replicate converged under `G9` (zero non-convergence, against 1.0% and 1.6% for `G6`/`G7`), and the power gain is the largest observed, which is at least consistent with (though does not itself establish) CR2 correcting real conservatism rather than papering over a different problem. This block did not test the null ($c_{bm} = 0$), so it does not resolve whether `G4`/`G9`'s Type I error is controlled; Section S11, next, checks that directly.

6 S11: is G4's Type I error controlled?

`G4` selects its half-life by AIC and then tests the interaction coefficient from the selected model, a post-selection inference problem: the same data drive both the selection and the test, which can in principle inflate Type I error beyond nominal. This had not been checked anywhere in this manuscript's development. Block S11 tests it directly, refitting `G1-G5` and `G9` at the null ($c_{bm} = 0$) at the Hybrid reference cell, with $n_{sim} = 2000$ for tighter precision on a rate expected near 0.05.

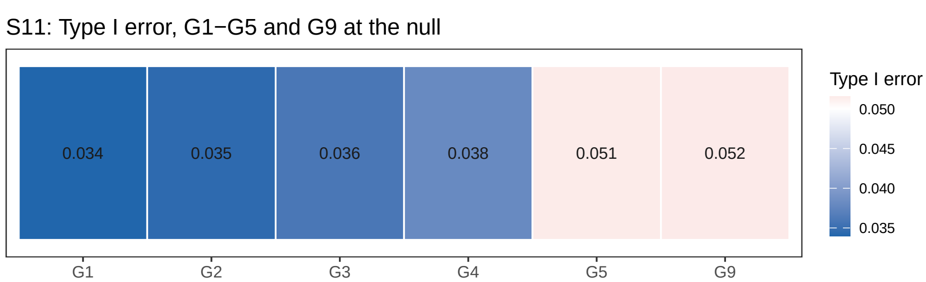

Figure 13: Block S11: empirical Type I error under the null for G1-G5 and G9 (diverging scale centered at nominal 0.05).

The concern does not materialize. `G4`'s Type I error is 0.038 (MCSE 0.004), conservative and in the same range as `G1-G3` (0.034-0.036), not inflated by the selection step; if anything it sits slightly closer to nominal than `G1/G3`. `G9` (`G4`+CR2) is 0.052, indistinguishable from nominal 0.05 (well under one MCSE), matching `G5`'s own 0.052. Non-convergence for `G4/G9` was negligible (0.05%, one replicate of 2000). Read together with Section S10's power results, this means `G9`'s status as the best-performing specification in Section S10 is not an artifact of an uncontrolled test: it wins on power while holding Type I error at nominal, the same standard `G3`+CR2 (`G8`) is held to. This does not generalize beyond the single cell tested; the post-selection inference risk that motivated this block is a property of the estimator, not of this specific cell, but only this cell was checked.

7 S12: Architecture A counterpart to Figures 1-3 (exploratory)

The main manuscript restricts every result except Block S8 to Architecture B (covariance moderation), the scope stated in Section 2.3. This section is exploratory material outside that scope: an Architecture A (mean moderation) counterpart to the main manuscript's Figures 1-3, generated on request for comparison. It is not part of this manuscript's evidence base and is not referenced by any claim in the main text.

Panels A and B (all nine specifications) are from a dedicated 30-cell simulation restricted to Architecture A, otherwise identical in design to the Architecture B simulation behind the main manuscript's Figures 1-2 (`21-run-tier1-hendrickson-g9-archA.R`, $n_{sim} = 250$, after a 100-rep smoke test passed sanity checks: no missing values, power spanning 0.02-0.98, mean non-convergence 0.08%, no cell above 2%). Panel C (`G1-G3` only, decay-shape axis) is from a second dedicated 30-cell simulation at the same $n_{sim} = 250$, matching the more extreme Weibull shape range ($k = 0.25, 0.5, 2.0, 4.0$) the main manuscript's Figure 3 uses (`24-run-decay-shape-sensitivity-archA.R`), rather than the shared production grid (`02-grid-summary.rds`, still at the earlier $k = 0.7, 1.0, 1.5$).

This is consistent with, and considerably sharper than, Block S8's finding (Section 3.7 of the main text) that the Exposure-weighted family's advantage is architecture-sensitive. At the CO cell ($N = 70$, $c_{bm} = 0.45$, $t_{1/2} = 1.0$), the Exposure-weighted family trails `G1/G2/G6/G7` by a similar margin under both architectures (`G4` 0.54 against `G1` 0.85 here, against `G4` 0.48 / `G1` 0.86 under Architecture B at the same cell). At Hybrid, the separation that favors Exposure-weighted under Architecture B collapses under Architecture A, but not by reversing: every specification is near ceiling there (`G1` 0.972, `G3` 0.964, `G4` 0.968, `G9` 0.964), a compression that likely reflects Architecture A's mean-moderation channel producing higher absolute power at this c_{bm} in a design with abundant on-drug information, leaving little room for any specification to separate from another. At OL+BDC the pattern does reverse, modestly: `G1` (0.928) edges out `G3/G4/G9` (0.900/0.864/0.888) by 3 to 6 points. Readers should treat this section as a first look, not a validated comparison: a proper cross-architecture comparison, including the full factorial grid and the sensitivity battery Block S8 alone does not cover, is the subject of the companion manuscript on DGP architectures[1] and the archived full-scope drafts.

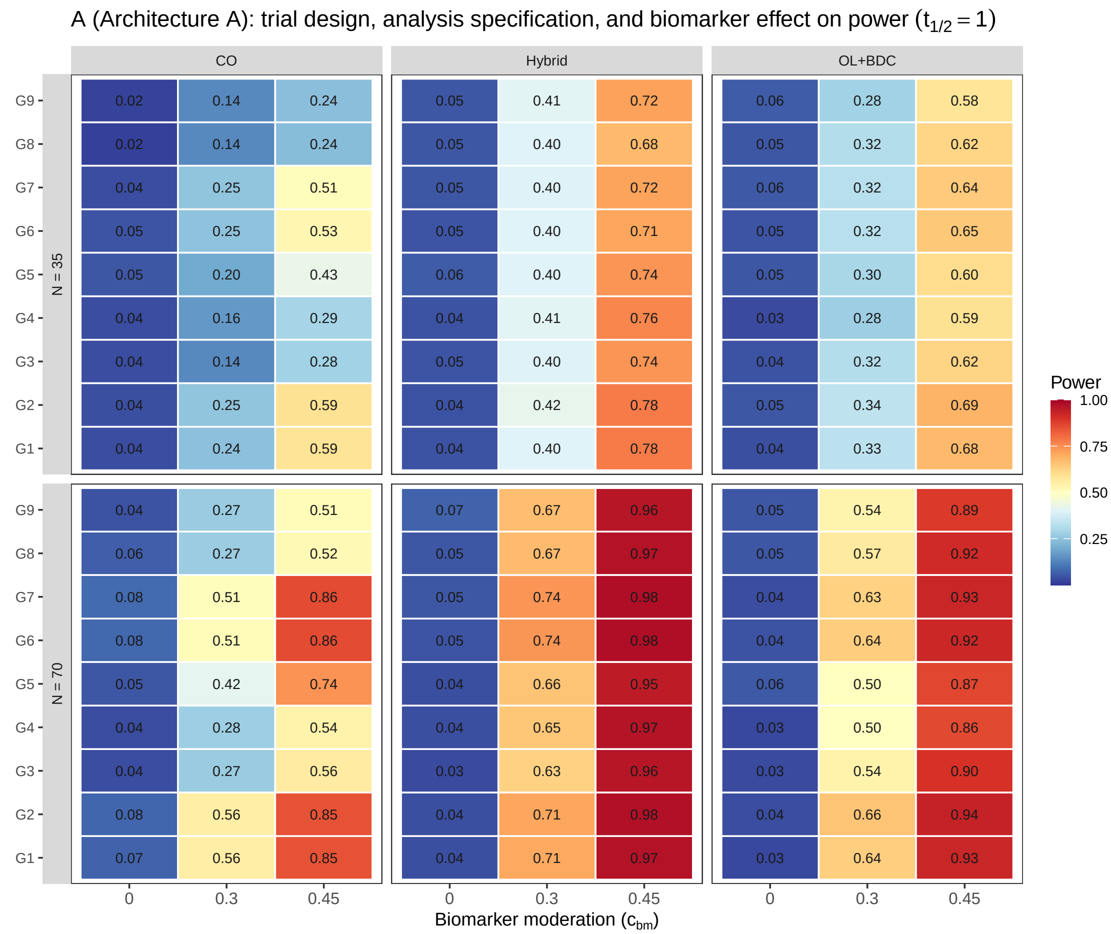

Figure 14: Architecture A counterpart to Figure 1: biomarker-effect axis, all nine G1-G9 specifications, power by trial design (columns), N (row blocks), and analysis specification (rows within block), across $c_{bm} \in \{0, 0.30, 0.45\}$ at $t_{1/2} = 1.0$ weeks, exponential DGP, Architecture A ($n_{sim} = 250$).

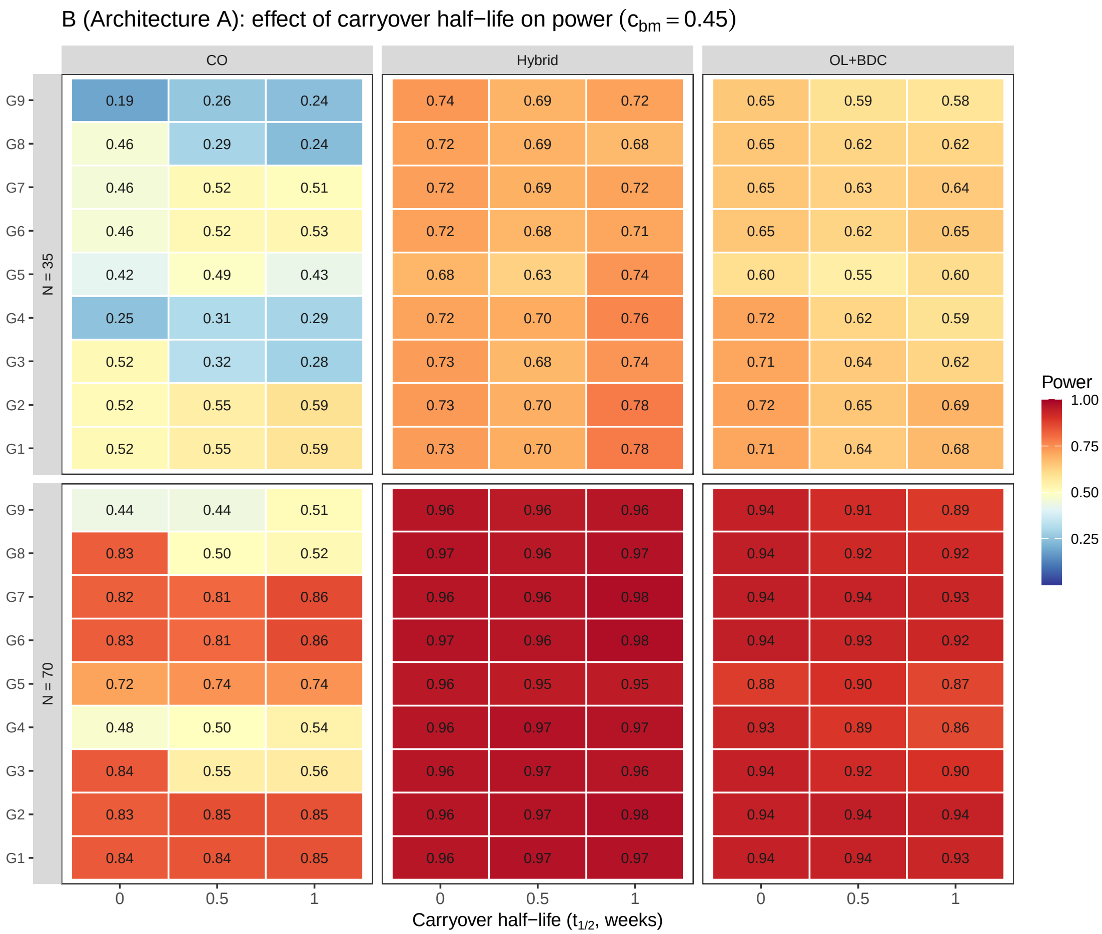

Figure 15: Architecture A counterpart to Figure 2: carryover-axis, all nine G1-G9 specifications, power by trial design (columns), N (row blocks), and analysis specification (rows within block), across $t_{1/2} \in \{0, 0.5, 1.0\}$ weeks at $c_{bm} = 0.45$, exponential DGP, Architecture A ($n_{sim} = 250$).

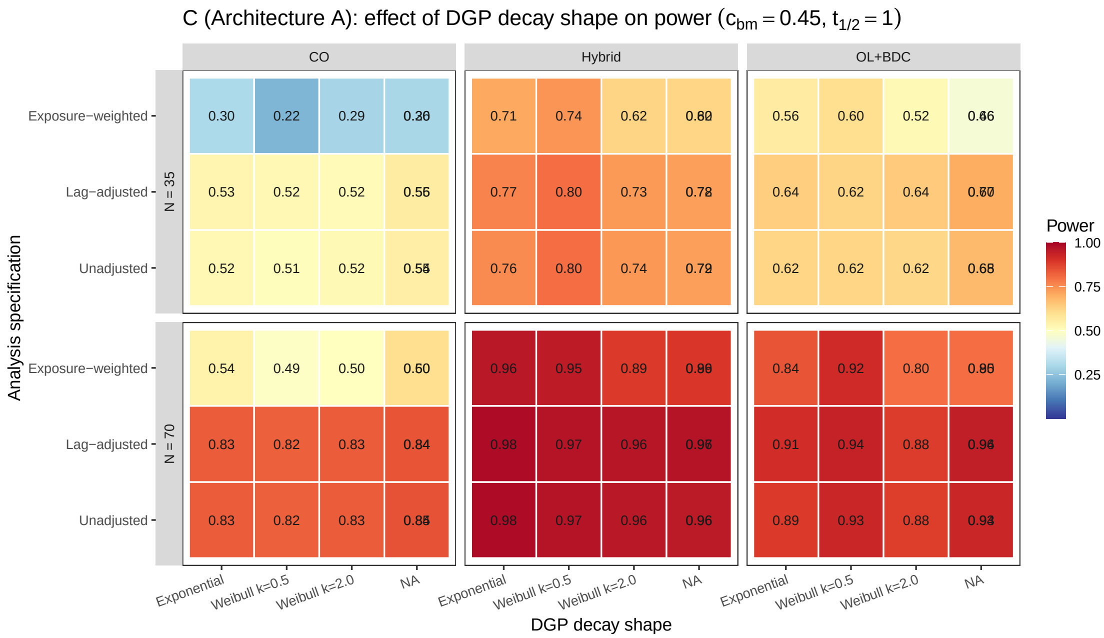

Figure 16: Architecture A counterpart to Figure 3: decay-shape axis, power by trial design (columns), N (row blocks), and analysis specification (rows within block), across the five DGP decay-shape levels (exponential, Weibull $k \in \{0.25, 0.5, 2.0, 4.0\}$) at $c_{bm} = 0.45$, $t_{1/2} = 1.0$ weeks, Architecture A ($n_{sim} = 250$).

1. pmsimstats team. Two architectures for simulating biomarker-treatment interactions in aggregated N-of-1 trials. Published online 2026.